Q projection op exercise

$X[32, 1024, 8192] \cdot W_Q[8192, 32, 256]$ on v2 4×2 slice

$BS = \frac{32 \cdot 1024}{4} = 8192$ so we're compute bound

$FLOPS = 2 \cdot 32 \cdot 1024 \cdot 8192 \cdot 32 \cdot 256 \approx 2.46 \cdot 10^{12}$

$$T_{comp} = \frac{2.46 \cdot 10^{12}}{8 \cdot 23 \cdot 10^{12}} = 13.4 \text{ ms}$$

This is close to the wall duration in the profile!

Worked Problems

1) Trace:

multiply_reduce_fusion.1

$\underbrace{\text{bf16}[4096, 8192]}_{W_1} \cdot \underbrace{\text{bf16}[8192]}_{In} \rightarrow \underbrace{\text{bf16}[4096]}_{Tmp}$

⇒ matmul

multiply_reduce_fusion

$\underbrace{\text{bf16}[8192, 4096]}_{W_2} \cdot \underbrace{\text{bf16}[4096]}_{Tmp} \rightarrow \underbrace{\text{bf16}[8192]}_{Out}$

⇒ matmul (feeding in result from last HLO op) all-reduce(bf16[1, 8192]) $\rightarrow$ bf16[1, 8192]

Running on a v5e-8 slice and replica-groups is 8, so sharded across all devices on contracting dim.

Computation:

$$W_1[I_{xy}, J] \cdot_J In[J] \rightarrow Tmp[I_{xy}]$$

$$W_2[K, I_{xy}] \cdot_I Tmp[I_{xy}] \rightarrow Out[K]\{U_{xy}\}$$

$$AllReduce_{xy}(Out[K]\{U_{xy}\}) \rightarrow Out[K]$$

$$\underbrace{bf16[32768, 8192]}_{W_1} \cdot \underbrace{bf16[8192]}_{In} \cdot \underbrace{bf16[8192, 32768]}_{W_2} \rightarrow bf16[8192]$$

This operation is essentially the up and down projection in an MLP where the F dim is sharded across all 8 devices.

2)